QUASI-COHERENT MODULES ON THE SIMPLICIAL TOTAL SITE ARE EXACTLY THE CARTESIAN ONES WITH QUASI-COHERENT COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the closing 'We omit the details' (source line 2890) covers the check that cartesianness makes the pulled-back presentation exact: pullback of modules is right exact, so the pulled-back free presentation remains a presentation of the pulled-back cokernel, and the cartesianness isomorphisms F(varphi) identify that cokernel with F_m componentwise; the constructions on both directions are written in full, only this routine verification is unexpanded; this lemma is a stated ingredient of card L46153 of this batch.

Editorial note. The label lemma-quasi-coherent-sheaf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Modules on Sites, the site-local definition of quasi-coherence. Modules on Sites chapter, \label{sites-modules-definition-site-local}. Lines 2819-2821: the sense of quasi-coherence used on both sides of the equivalence.
- Pullbacks of quasi-coherent modules are quasi-coherent. Modules on Sites chapter, \label{sites-modules-lemma-local-pullback}. Lines 2829-2832: the restrictions F_n = g_n^*F are quasi-coherent in the forward direction.
- Coverings of C_n versus coverings of C_total. Same chapter, Situation \ref{situation-simplicial-site} and the construction of the site C_total. Lines 2842-2844: a covering of C_total over an object of C_n is a covering of C_n by construction.
- Morphism of sites: stalks and coverings (case A). Sites chapter, \label{sites-lemma-morphism-of-sites-covering}. Line 2868: closes the cartesianness argument in case A.

Source reference. Original lemma 08H2: sources/stacks-chow/tags.txt, source lines 10765–10765. Prerequisite pointer invoked at proof line(s) 2870; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Prerequisite pointer invoked at proof line(s) 2822; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03DO: sources/stacks-chow/tags.txt, source lines 4427–4427. Prerequisite pointer invoked at proof line(s) 2830; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 2818; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}_{total}$. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Then $\mathcal{F}$ is quasi-coherent in the sense of Modules on Sites, Definition 03DL if and only if $\mathcal{F}$ is cartesian and $\mathcal{F}_n$ is a quasi-coherent $\mathcal{O}_n$-module for all n.*

Proof. Assume $\mathcal{F}$ is quasi-coherent. Since pullbacks of quasi-coherent modules are quasi-coherent (Modules on Sites, Lemma 03DO) we see that $\mathcal{F}_n$ is a quasi-coherent $\mathcal{O}_n$-module for all n. To show that $\mathcal{F}$ is cartesian, let U be an object of $\mathcal{C}_n$ for some n. Let us view U as an object of $\mathcal{C}_{total}$. Because $\mathcal{F}$ is quasi-coherent there exists a covering $\{U_i \to U\}$ and for each i a presentation

$$\bigoplus\nolimits_{j\in J_i} \mathcal{O}_{\mathcal{C}_{total}/U_i} \to \bigoplus\nolimits_{k\in K_i} \mathcal{O}_{\mathcal{C}_{total}/U_i} \to \mathcal{F}|_{\mathcal{C}_{total}/U_i} \to 0$$

Observe that $\{U_i \to U\}$ is a covering of $\mathcal{C}_n$ by the construction of the site $\mathcal{C}_{total}$. Next, let V be an object of $\mathcal{C}_m$ for some m and let $V \to U$ be a morphism of $\mathcal{C}_{total}$ lying over $\varphi : [n] \to [m]$. The fibre products $V_i = V \times_U U_i$ exist and we get an induced covering $\{V_i \to V\}$ in $\mathcal{C}_m$. Restricting the presentation above to the sites $\mathcal{C}_n/U_i$ and $\mathcal{C}_m/V_i$ we obtain presentations

$$\bigoplus\nolimits_{j\in J_i} \mathcal{O}_{\mathcal{C}_m/U_i} \to \bigoplus\nolimits_{k\in K_i} \mathcal{O}_{\mathcal{C}_m/U_i} \to \mathcal{F}_n|_{\mathcal{C}_n/U_i} \to 0$$

and

$$\bigoplus\nolimits_{j\in J_i} \mathcal{O}_{\mathcal{C}_m/V_i} \to \bigoplus\nolimits_{k\in K_i} \mathcal{O}_{\mathcal{C}_m/V_i} \to \mathcal{F}_m|_{\mathcal{C}_m/V_i} \to 0$$

These presentations are compatible with the map $\mathcal{F}(\varphi) : f_\varphi^*\mathcal{F}_n \to \mathcal{F}_m$ (as this map is defined using the restriction maps of $\mathcal{F}$ along morphisms of $\mathcal{C}_{total}$ lying over φ). We conclude that $\mathcal{F}(\varphi)|_{\mathcal{C}_m/V_i}$ is an isomorphism. As $\{V_i \to V\}$ is a covering we conclude $\mathcal{F}(\varphi)|_{\mathcal{C}_m/V}$ is an isomorphism. Since V and U were arbitrary this proves that $\mathcal{F}$ is cartesian. (In case A use Sites, Lemma 08H2.)

Conversely, assume $\mathcal{F}_n$ is quasi-coherent for all n and that $\mathcal{F}$ is cartesian. Then for any n and object U of $\mathcal{C}_n$ we can choose a covering $\{U_i \to U\}$ of $\mathcal{C}_n$ and for each i a presentation

$$\bigoplus\nolimits_{j\in J_i} \mathcal{O}_{\mathcal{C}_m/U_i} \to \bigoplus\nolimits_{k\in K_i} \mathcal{O}_{\mathcal{C}_m/U_i} \to \mathcal{F}_n|_{\mathcal{C}_n/U_i} \to 0$$

Pulling back to $\mathcal{C}_{total}/U_i$ we obtain complexes

$$\bigoplus\nolimits_{j\in J_i} \mathcal{O}_{\mathcal{C}_{total}/U_i} \to \bigoplus\nolimits_{k\in K_i} \mathcal{O}_{\mathcal{C}_{total}/U_i} \to \mathcal{F}|_{\mathcal{C}_{total}/U_i} \to 0$$

of modules on $\mathcal{C}_{total}/U_i$. Then the property that $\mathcal{F}$ is cartesian implies that this is exact. We omit the details. □